

Machine Learning Analysis Report

Project: Applied Machine Learning — Model Design, Training, and Performance Evaluation

Task: Supervised binary classification of adult census income

Notebook: `modeling.ipynb`

Overview

This project addresses a supervised binary classification problem: predict whether a person earns more than \$50,000 per year from demographic and employment attributes. The data are the Adult (Census Income) dataset from the UCI Machine Learning Repository (Becker & Kohavi, 1996; Kohavi, 1996), available at <https://archive.ics.uci.edu/dataset/2/adult>. The primary model is logistic regression implemented in a scikit-learn pipeline; a random forest with the same preprocessing is used as a nonlinear comparison.

Dataset Description

The Adult dataset is derived from 1994 U.S. Census Bureau data and is a standard public benchmark for income classification (Becker & Kohavi, 1996). After combining the official training and test files, the working table has 48,842 rows and 15 columns. Features include age, work class, education (label and numeric years), marital status, occupation, relationship, race, sex, capital gain and loss, hours worked per week, native country, and a census sampling weight (`fnlwgt`). The target variable is `income`, coded as `<=50K` or `>50K`. About 76.1% of records are `<=50K` and 23.9% are `>50K`. Missing values occur in `workclass` (2,799), `occupation` (2,809), and `native-country` (857).

Modeling Approach

Data were split with a stratified 80/20 train-test partition (`random\_state=42`) so that class proportions were preserved and evaluation used records the model had not seen. `fnlwgt` was dropped because it is a survey weight rather than a personal attribute. `education` was dropped in favor of `education-num` to avoid duplicating the same information. Numeric features were median-imputed and standardized; categorical features were imputed with the most frequent category and one-hot encoded. Transformers were fit only on the training fold inside a `Pipeline` so that preprocessing could not leak test information (Pedregosa et al., 2011).

Logistic regression was chosen as the primary model because the task is binary classification, the algorithm yields class probabilities, and coefficients can be inspected after encoding (Hosmer et al., 2013). `class\_weight="balanced"` was used so that the minority `>50K` class was not ignored during training. A random forest with 100 trees and `max\_depth=20` was trained with the same

preprocessor as a comparison that can capture interactions without an explicit linear log-odds assumption.

Accuracy is a poor sole metric under class imbalance, because a classifier that always predicts ` $\leq 50K$ ` would already be correct on about 76% of cases. Precision, recall, and F1-score for the ` $>50K$ ` class measure whether high-income cases are recovered without flooding the prediction with false positives (Sokolova & Lapalme, 2009). ROC-AUC summarizes ranking quality across decision thresholds (Fawcett, 2006). Logistic regression assumes a linear relationship between features and the log-odds of the positive class, independent observations, and that one-hot encoded categories are adequately represented. Those assumptions are convenient for interpretation but are only approximations of census-taking and labor-market processes.

Results

All figures below refer to the held-out test set of 9,769 rows (2,338 ` $>50K$ `).

Model	Accuracy	Precision ($>50K$)	Recall ($>50K$)	F1 ($>50K$)	ROC-AUC
Logistic regression	0.8069	0.5653	0.8366	0.6747	0.9037
Random forest	0.8238	0.5924	0.8456	0.6967	0.9192

The logistic model reached 80.7% accuracy and an F1 of 0.67 on the high-income class, with recall 0.84 and precision 0.57. It therefore finds most ` $>50K$ ` cases but also labels many ` $\leq 50K$ ` people as high-income. The random forest improved modestly on every reported metric (accuracy 0.82, F1 0.70, ROC-AUC 0.92). Confusion-matrix heatmaps and ROC curves in `modeling.ipynb` show the same pattern: both models rank well (ROC-AUC above 0.90), while the operating point induced by class weighting favors recall over precision.

Subgroup evaluation for logistic regression on the test set showed a higher base rate of ` $>50K$ ` for men (30.4%, $n = 6,480$) than for women (11.2%, $n = 3,289$). Recall for ` $>50K$ ` was 0.87 for men and 0.68 for women; F1 was 0.69 versus 0.61. White respondents ($n = 8,348$) had recall 0.85, while Black respondents ($n = 944$) had recall 0.65, despite a lower positive base rate in the latter group. These gaps are descriptive of this fitted pipeline; they are not a causal analysis of wages.

The largest-magnitude logistic coefficients associated capital gain and being in a married-civilian or married-armed-forces household with higher log-odds of ` $>50K$ `, and occupations such as private household service with lower log-odds. Coefficients are associations after controlling for the other encoded features, not proof of causation.

Interpretation for a Non-Technical Audience

The model tries to guess, from census-style facts such as age, job type, education, and hours worked, whether someone is in the higher income group (over \$50,000

in 1994 dollars). On people it had not seen during training, it was right about four times out of five. When someone truly was in the higher-income group, the logistic model usually flagged them (about 84% of those cases). However, many of the people it flagged as high-income were not, so the “high income” label should not be treated as a reliable screening decision on its own. A second model (a random forest) was slightly more accurate, but the improvement was small. Performance also differed by sex and race: the model more often missed high-income women and Black respondents than high-income men and White respondents. In short, the notebook shows a useful statistical pattern in historical census data, not a fair or complete picture of a person’s earning potential.

Limitations and Potential Bias

A central limitation is the 1994 snapshot: occupations, wages, and household structures have changed, so the fitted mapping from features to the \$50,000 threshold does not automatically transfer to current populations. Class weighting raises recall for ` $>50K$ ` at the expense of precision; a different threshold or cost of errors could reverse that tradeoff. Logistic regression cannot represent complex interactions unless they are engineered as features, which is why the random forest was included as a check.

A potential source of bias is that sex and race are both recorded in the data and were used as model inputs. Historical census and labor-market processes already encode unequal access to high-paying work, so a model that predicts income from those fields can reproduce and operationalize that inequality (Mehrabi et al., 2021). Missing work class and occupation may also be non-random. One mitigation already taken in this project is to report recall and F1 by sex and race instead of only overall averages, which makes disparate error rates visible. A further step would be to exclude protected attributes from the feature set, audit residual disparity on those attributes, and refuse to use the scores for hiring, lending, or eligibility decisions without a documented fairness review.

References

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